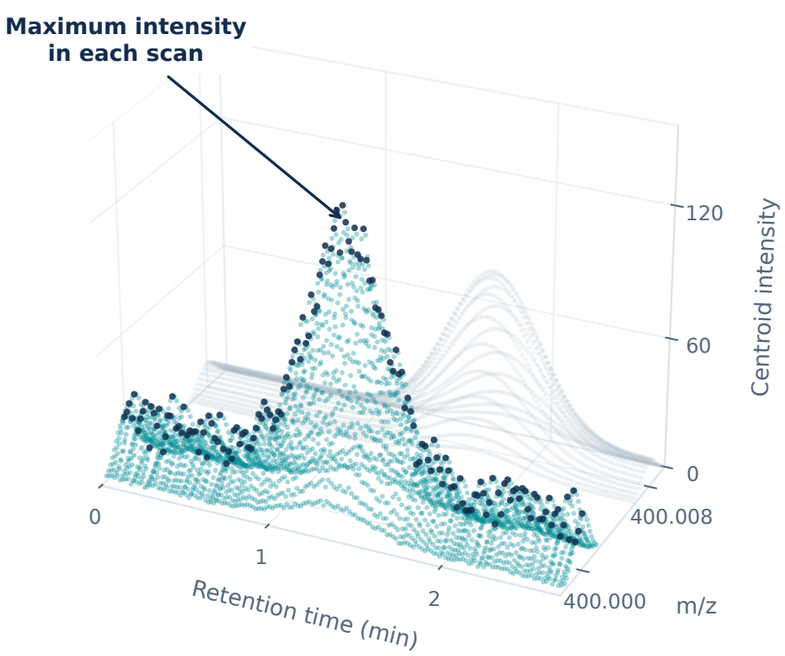
Estimate trace noise from fluctuations around a smooth signal

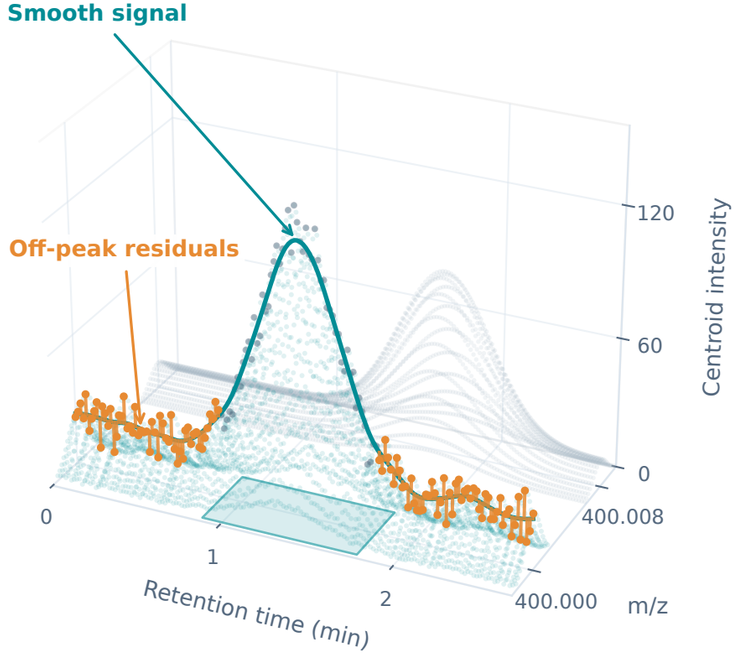
Default EIC check: use raw − smoothed residuals outside the candidate's peak bounds.

1 Read a trace from the cloud



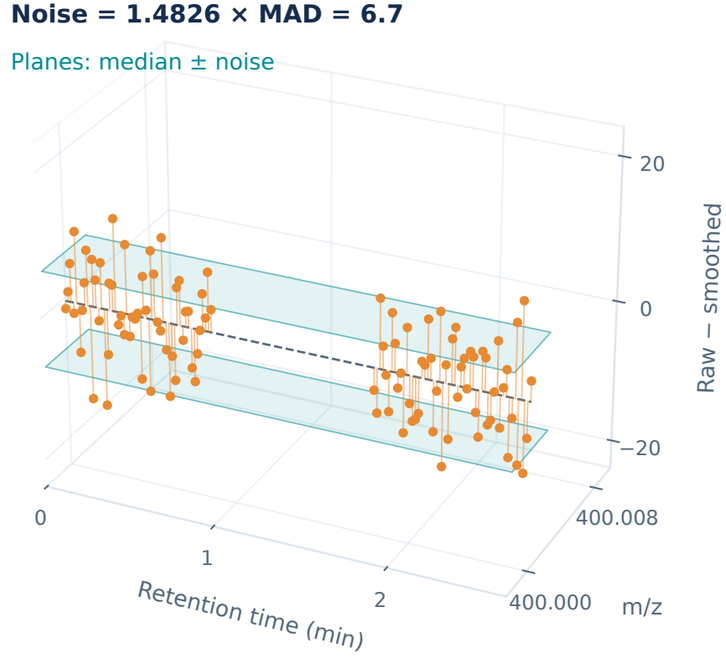
Select the ion's m/z window.
Keep the strongest centroid in each scan.

2 Measure off-peak fluctuations



Orange: raw − smooth outside peak bounds.
Tinted base: candidate interval, excluded here.

3 Measure robust noise spread



Robust spread of the selected residuals.
Default reference = max(6.7, cell floor 4.0).

Same ion and time window in every panel • Panel 3 expands the residual scale • MAD = median absolute deviation from the median

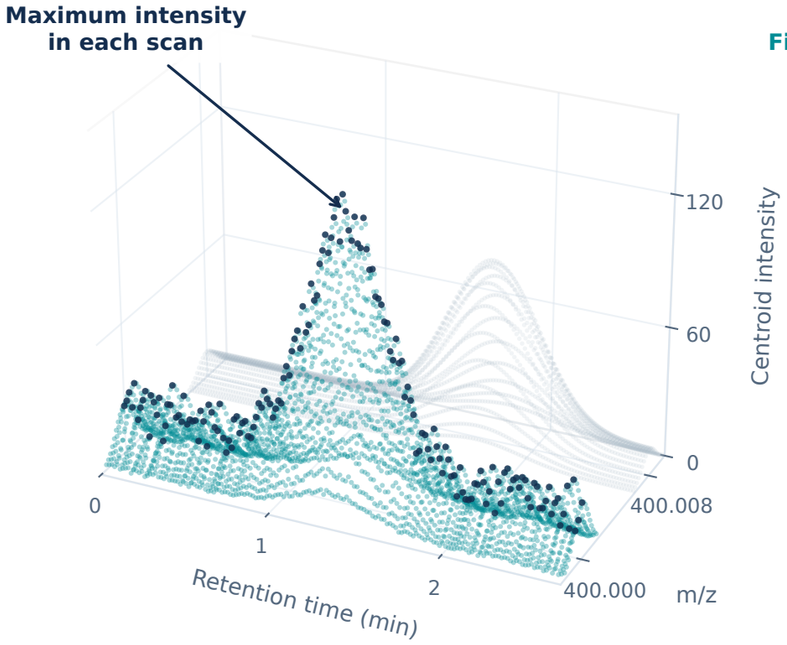
Extract an ion trace → subtract its smooth trend → compare trace noise with the local cell floor.

Synthetic data • exact eic_shape calculation • default eic_noise=max • illustrative cell floor = 4.0

Measure a trace's own scan-to-scan fluctuations

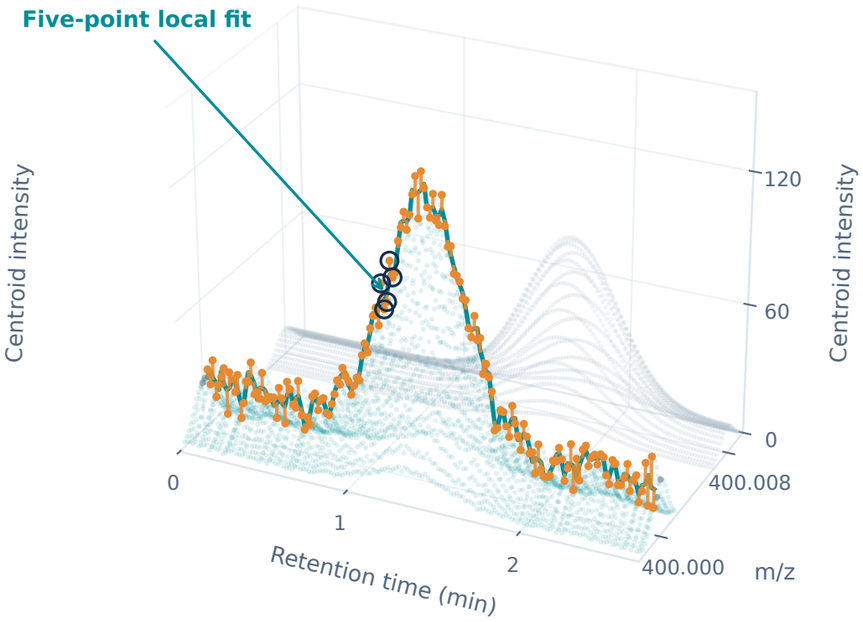
Optional own mode: a five-point local quadratic follows the peak; residual spread estimates its noise.

1 Read a trace from the cloud



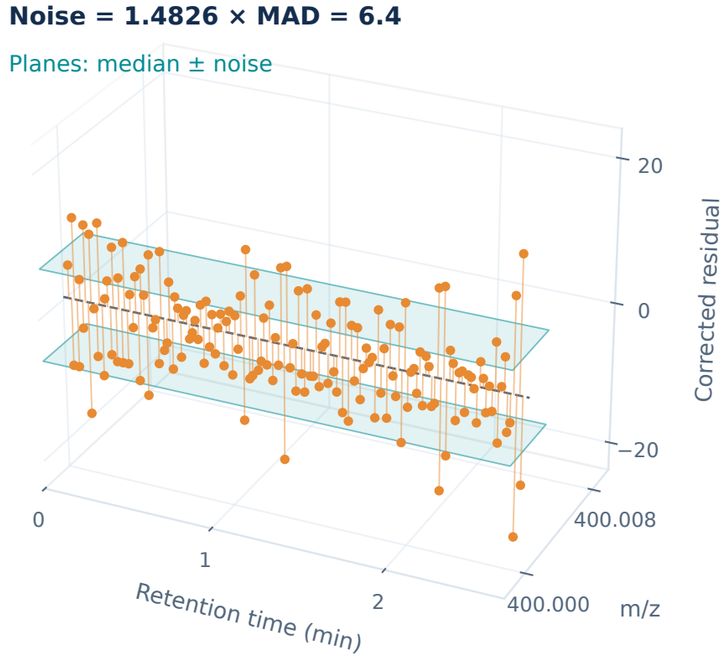
Select the ion's m/z window.
Keep the strongest centroid in each scan.

2 Fit five neighbouring scans



Fit the local quadratic, including the centre.
Keep only five-scan stretches with signal.

3 Measure robust noise spread



Correct residuals by √(35/18), then take MAD.
At least 10 valid residuals are required.

Same ion and time window in every panel • Panel 3 expands the residual scale • MAD = median absolute deviation from the median

Extract an ion trace → fit each five-scan neighbourhood → estimate robust residual spread.

Synthetic data • exact own_noise calculation • optional eic_noise=own; also used by merge_test=own